

AMTI (NMTTC) - 2017
RAMANUJAN - FINAL - INTER LEVEL

1. A, B, C are three points on a circle. The distance of C from the tangents at A and B to the circle are a and b respectively. If the distance of C from the chord AB is c, show that c is the geometric mean of a and b.
2. Find all integer solutions to the equation $x^3 + (x + 4)^2 = y^2$.
3. Two right angled triangles are such that the incircle of one triangle is equal in size to the circum circle of the other. If Δ_1 is the area of the first triangle and Δ_2 , the area of the second triangle, show that $\frac{\Delta_1}{\Delta_2} \geq 3 + 2\sqrt{2}$.
4. (a) Find the maximum value k for which one can choose k integers from 1, 2, ..., 2n so that none of the chosen integers is divisible by any other chosen integer.
(b) $F(x)$ is a polynomial of degree 2016 such that all the coefficients are non negative and none exceed $F(0)$. Show that the coefficient of x^{2017} in $(F(x))^2$ is at most $\frac{F(1)^2}{2}$.
5. (a) $n \geq 3$ and $a_1, a_2, \dots, a_n$ are different positive integers. Given that, except the first and the last, each one is a harmonic means of its immediate neighbors. Show that none of the given integers is less than $n - 1$.
(b) Show that the shortest side of a cyclic quadrilateral with circumradius 1 is at most $\sqrt{2}$.
6. C_1, C_2, C_3 are circles with radii 1, 2, 3 respectively, touching each other as shown. Two circles can be drawn touching all these three circles. Find the radii of these two circles.
